

Signal-to-noise v/s Contrast-to-noise

- Signal-to-noise ratio (SNR)
- SNR provides a means to estimate the precision with which the signal S is measured
 - S: signal (or measurement variable)
 - σ : standard deviation of its measurement

$$SNR = \frac{S}{\sigma}$$

When is it possible to have excellent SNR but poor CNR ..?

To discriminate two signals S_1 and S_2 we need more than just good signal to noise ratio. The ability to discriminate the two is assessed using the contrast to noise ratio...

Contrast-to-noise ratio (CNR)

S_1 and S_2 : two signals (or measurement variable) of two different tissues,

σ : standard deviation of their independent measurement (assumed here to be identical and statistically independent)

$$CNR = \frac{S_1 - S_2}{\sigma}$$

CNR provides a means to estimate the precision with which S_1 can be discriminated from S_2

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Optimizing CNR ?

- Optimizing contrast = choice of experimental parameters (e.g. protocol) to maximize the difference in two tissue signals S_1 and S_2 .
 - complex and empirical procedure
 - some effects can be predicted/calculated, if the signal behavior can be modeled.

Error propagation calculation

Let the signal S be a function $S(k, t)$
 k is a tissue property (signal decay rate)
 t an experimental parameter (such as time).

Approach:

1. Determine dS/dk
2. Find t_0 where dS/dk is maximal by taking derivative rel. to t

Example: $S(k, t) = S_0 e^{-kt}$

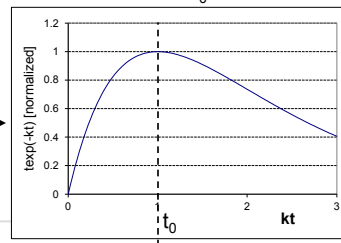
$$\frac{dS(k, t)}{dk} = -S_0 t e^{-kt}$$

Maximum is where derivative with respect to t is zero

$$\begin{aligned} \frac{d}{dt}(-S_0 t e^{-kt}) &= 0 \\ &= -S_0 e^{-kt} + S_0 k t e^{-kt} \\ &= -S_0 e^{-kt} (1 - kt) = 0 \\ &\quad \boxed{t_0 = 1/k} \end{aligned}$$

For an exponentially decaying signal, the optimal time of measurement is equal to 1/decay rate

How critical is the choice of t_0 ?



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